Apartment 50: Paper 002

AuthorityEffect: None. This is fictional examination staging, not a historical quotation or a source of mathematical authority.

Landau asks for the calculation and both directions of the proof. Woland asks which claim the checker can establish. Behemoth changes an edge bit, rehashes a false receipt, and asks whether anyone notices. Their task in this companion is **signed certificates, honest search frontiers, and the restricted integrated-path obstruction**.

Begin with the study guide, attempt the quiz, and only then open the separate worked instructor key. Levels 0-7 move from a declared sign equation to the all-length weight gap and a publication evidence audit. A symbol names a role only after its interpretation is declared. Walk arrows do not attach higher-dimensional cells. A receipt binds its stated check and grants no authority to extend Genesis or enlarge the physical claim.

The examiner's standard is a concrete calculation, an explicit counterexample, or a precisely identified missing obligation. The manuscript governs the answers. The staging is literary; the score is an instructional convention.

Study guide: PFP II

AuthorityEffect: None. Start with the fictional Apartment 50 preface. This guide prepares the 14-problem quiz; complete an attempt before opening the separate worked instructor key. All required readings and checks are included in this deposit.

What you need and what you should be able to do

Prerequisites are addition modulo two, finite graph walks, matrix multiplication, exact fractions, a spanning-tree argument, factorials, and the power series for the exponential. For the Fock example, use the manuscript's explicit occupation and hopping rules; no prior many-body simulation experience is required. Review those rules before attempting the integration and length bounds.

By the end, produce and independently check both witness types; derive the matrix adapter; distinguish a complete classifier from an unfinished enumeration; replay a contiguous receipt chain; and explain exactly which pairing operation the Fock example excludes.

Notation and the objects being checked

Object	Meaning	Typical mistake
$c_e \in \mathbb{Z}_2$	An edge's endpoint constraint	Treating XOR as rational addition
α_v	A vertex bit; sign $(-1)^{\alpha_v}$	Assuming the chosen root removes candidates from the implementation
Odd closed walk	Indexed traversals, closure, parity one	Checking parity without adjacency or closure
Exact matrix	Integers or `Fraction` values in a fixed basis	Silently rounding a floating input
Presentation key	Full declared context for retrieval and replay	Matching only the Hamiltonian digest
Receipt frontier $[s, t)$	The suffix outside the recorded checks	Treating an unfinished suffix as a negative proof
Integrated rooted path	A state sequence whose jump times were integrated out	Equating finite state count with finite sequence count

An arrow records a traversal in a walk. A graph edge does not attach a two-dimensional cell. A bracket-wire encoding, if used, needs its own declared interpretation and parsing rules.

Route through Levels 0-7

Level	Read in the manuscript	Practice	Completion check
0	Sections 1-2	Q1	State the carrier and exact move grammar
1	Section 2	Q2-Q3	Check a gauge and construct indexed obstructions
2	Section 2	Q4-Q5	Separate coefficients and prove both directions
3	Sections 1 and 3	Q6-Q7	Compute DHD and prove exponential positivity
4	Section 4	Q8-Q10	Account for cursor, batch, key, and prior coverage
5	Section 5	Q11	Check an actual pairing map, not just its sum
6	Section 6	Q12-Q13	Derive the target and cover every length class
7	Sections 7-8 and the claim ledger	Q14	Match an exact claim to its evidence and limit

1. Local equations and the global witness

Practice first on a tree: root one vertex at zero and propagate its bits. There is no cycle constraint to violate. Add a closing edge and compare the required endpoint bit with the propagated bit. A disagreement supplies an odd closed walk through the tree.

For a gauge, check every edge equation, including loops and repeated endpoints. For an obstruction, check every edge ID and endpoint, continuity between steps, final closure, and the XOR of its bits. Reversing a traversal does not change a binary edge bit. Give each parallel edge its own identity.

Then change coefficients. Rational reports on an oriented cycle require a telescoping sum of zero. Modulo-two parity can vanish while that rational sum does not. Keep the coefficient system and allowed report repairs in the statement.

2. The matrix bridge

The adapter assigns a bit one to a positive nonzero off-diagonal entry and zero to a negative one. This bit is the constraint needed to make the transformed matrix entry negative; the graph sign $(-1)^{c_e}$ follows that convention. Zero entries impose no equation.

Compute an entry of DHD as $D_{ii}H_{ij}D_{jj}$ before applying a general rule. A successful gauge proves stoquasticity under those diagonal signs in the declared basis. Other basis changes require a different grammar.

For exponential positivity, shift the diagonal so $B = aI - H'$ is entrywise nonnegative. Powers of B remain nonnegative, and the convergent power series gives the result for nonnegative β. This establishes matrix-entry positivity; sampling cost and observable-estimator variance require additional analysis.

3. Complete classification and incomplete search

The graph classifier is linear in the finite input size. Exhaustive enumeration is retained to exercise a protocol for future grammars; its worst-case candidate count is exponential.

Search status	Evidence required	What follows
`GAUGE_VERIFIED`	A gauge passes all equations	That gauge works in this grammar
`UNKNOWN`	The recorded segment has no valid gauge and a suffix remains	Continue only with a verified prior prefix
`RESTRICTED_OBSTRUCTION`	Replayed exhaustive coverage and a checked odd walk	No vertex gauge exists for this graph

The direct graph classifier can supply an odd walk without exhaustive enumeration. The search function issues its restricted terminal status only after its own finite grammar is exhausted.

For the three-vertex example the implementation has eight candidates, indexed zero through seven, even when a proof fixes a root. Track `start_index`, ordered checks, `next_index`, and `stop_index`. The cursor advances by the full consumed batch, not necessarily to one past the first valid candidate.

A digest binds bytes; replay checks their asserted equations. Continuation requires a contiguous chain of UNKNOWN segments beginning at zero, with the

same graph and presentation digests. Changing the basis, grammar, estimator, adapter, or another key field changes the declared context. A retrieved bit vector must be checked again even after a full-key match.

4. Cancellation and the infinite family of lengths

A pairing is a map on a closed paired subset. Check involution, absence of fixed points, opposite signs, and equal magnitudes. Separate unpaired survivors from that subset. A nonnegative total does not by itself supply a pairing.

For the Fock example, keep the six-mode order fixed. Count occupied modes strictly between a hop's endpoints for its sign and doubly occupied sites for its energy. A constant-energy three-hop sequence has magnitude $e^{-\beta E}\beta^3/3!$.

The proof has two different parts: exact enumeration establishes the special facts for length three, while inequalities cover every other length. For $n \geq 4$, use

$$\frac{\beta^{n+1}/(n+1)!}{\beta^n/n!} = \frac{\beta}{n+1} < 1 \quad (\beta = 1/4).$$

This controls an individual sequence. Counting and summing many sequences is a separate task.

For $x \in [0, 1]$, odd truncations of the alternating series for e^{-x} are lower bounds and even truncations are upper bounds. Compare rational enclosures instead of rounded decimals. The target lies above every long-path upper bound and below every relevant positive short-path lower bound, excluding an equal-magnitude partner.

5. Replay the package and describe the evidence

From the package root, use the README commands. On Windows use `py -3.12`; elsewhere substitute your Python 3.12 interpreter. The runtime checkers use the standard library. Tests require pytest; notebook replay additionally requires nbclient, nbformat, and ipykernel.

Replay	Supports	Does not establish
`code/tests/test_sign_graph.py`	Input and signed-witness checks	A new graph theorem
`code/tests/test_manuscript_examples.py`	Displayed matrix and 2-then-1 search trace	General physical adapter validity
`code/tests/test_frontier_search.py`	Coverage, ordering, key, and snapshot controls	Sampling speedup
`code/exact_fock_gap.py` and its test file	Exact counts and rational gap checks	Exclusion of every representation

Replay	Supports	Does not establish
Native-adapter tests	Adapter behavior; fallback if no executable exists	A native build when binary discovery skips
Fresh notebook replay	The deposited notebook executes under the recorded environment	Independent experimental data
`verify_deposit.py` and release ZIP check	File census, hashes, archive identity	Truth of a theorem

The claim ledger maps the paper's statements to proofs and named checks. Consult the current dated review receipt for actual execution, versions, and skips; historical counts keep their original scope.

Finish with a defense

Complete the quiz, mark it against the separate key, and write down the reason for each error: wrong carrier, wrong coefficients, invalid witness, missing coverage, or overextended claim. Rework the associated example before the oral defense. The score and mastery target are teaching conventions; no learner study or automatic authority is claimed.

PFP II Landau theoretical minimum: worked instructor key

Worked instructor key: PFP II

AuthorityEffect: None. Attempt the quiz first. The key explains the expected reasoning and a 100-point rubric. It is an educational companion to the scoped proofs and implementation, not an additional publication claim.

Q1. Declaration (6 points)

The carrier is the ordered finite vertices (a, b, c) with the three indexed undirected edges specified in the quiz. Coefficients are $\mathbb{Z}_2$; allowed moves assign one sign to each vertex, equivalently one bit α_v. Acceptance requires $\alpha_u \oplus \alpha_v = c_e$ on every edge. Bit zero requires equal bits and bit one opposite bits.

A name or drawing supplies no face, orientation-dependent rational report, or physical Hamiltonian. A bracket encoding needs an interpretation before it denotes any of these objects.

Credit: carrier and edge identity 2; coefficients and moves 2; acceptance equation and no undeclared structure 2.

Q2. All gauges (6 points)

With $\alpha_a = 0$, the first edge forces $\alpha_b = 1$, and the second forces $\alpha_c = 0$. The closing edge checks $0 \oplus 0 = 0$. The three equations are $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, and $0 \oplus 0 = 0$.

Without root-fixing, the gauges are 010 and 101. XORing two valid assignments gives a bit constant on each connected component, so these are the only two on this connected graph. Each additional consistent connected component supplies an independent global flip. In general a consistent graph with k components has 2^k gauges, including the empty graph's one empty assignment.

Credit: rooted calculation and all edge checks 3; two gauges and component freedom 3.

Q3. Indexed obstructions (6 points)

For the modified triangle, use `((0,a,b),(1,b,c),(2,c,a))`. It is continuous and closed, and its parity is $1 \oplus 1 \oplus 1 = 1$. XORing the three endpoint equations would give zero on the left and one on the right.

For one vertex a, a parity-one edge aa has witness `((0,a,a),)` and asks $\alpha_a \oplus \alpha_a = 1$. For two vertices, let edge 0 join a, b with parity zero and edge 1 join them with parity one. The witness `((0,a,b),(1,b,a))` has parity one. Distinct edge IDs are essential: the same undirected edge traversed out and back contributes its bit twice and cannot be this obstruction.

Removing loops excludes a valid declared input and changes the grammar.

Credit: triangle witness and contradiction 2; loop 2; parallel-edge witness and grammar distinction 2.

Q4. Rational near miss (8 points)

The binary parity of $(1, 1, 0)$ is zero. Its rational cycle sum is two. An exact potential would require

$$(p_b - p_a) + (p_c - p_b) + (p_a - p_c) = 0,$$

so these rational reports are not exact.

If the changes are $\delta_0, \delta_1, \delta_2$, their sum must be -2. The triangle inequality gives $\sum_i |\delta_i| \geq |\sum_i \delta_i| = 2$. Change the closing report from zero to -2, leaving reports $(1, 1, -2)$. The potential $(p_a, p_b, p_c) = (0, 1, 2)$ attains this repair with cost two. No two-dimensional filling was needed.

Credit: parity versus rational sum 2; telescoping obstruction 2; lower bound 2; attaining reports and potential 2.

Q5. Balance theorem and construction (10 points)

Necessity. XOR the endpoint equations around any closed walk, counting occurrences with multiplicity. Every arrival matches a departure around the closed walk, so the vertex terms cancel, including for a loop or a repeated edge. The edge XOR must be zero.

Sufficiency. Choose a root in each component and set its bit to zero. For each other vertex, define its bit as the parity of a root-to-vertex path. Any two such paths give a closed walk by reversing the second, so their parities agree. Append an edge uv to a root-to-u path; path independence proves $\alpha_v = \alpha_u \oplus c_{uv}$. This proves every edge equation. Isolated vertices are free; an odd loop or disagreeing parallel edges contradict the premise directly.

Breadth-first construction stores adjacency, assigned bits, and tree parents. Process each vertex and edge a bounded number of times. The first conflicting edge

plus the tree paths supplies a closed odd walk; if none conflicts, return the assignment. Producing that one walk costs at most linear extra work. Independent verifiers check the returned witness. The resulting $O(|V|+|E|)$ algorithm is the baseline; enumerating $2^{|V|}$ assignments exercises receipts rather than improving this classification algorithm.

Credit: necessity 3; sufficiency with path independence and components 4; constructive witnesses and linear versus exponential baseline 3.

Q6. Matrix (8 points)

The graph bits for ab, bc, ca are $1, 1, 0$, because the first two matrix entries are positive and the third negative. Entrywise multiplication gives

$$DHD = \begin{pmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}.$$

Multiplication by signs leaves zero unchanged, so a zero off-diagonal entry imposes no constraint. The exact adapter refuses floats, nonsquare arrays, and nonsymmetric arrays with an input-contract error; this is not an odd-walk proof for a well-formed instance.

An all-positive triangle excludes diagonal vertex signs in the fixed basis. It does not exclude an unrestricted global eigenbasis: every finite Hermitian matrix has a unitary diagonalization. Finding and using that basis can be computationally costly.

Credit: bits and computed matrix 4; zero and input refusals 2; transformation boundary 2.

Q7. Exponential (10 points)

Choose $a \geq \max_i H'_{ii}$. Then $B = aI - H'$ is entrywise nonnegative: its off-diagonal entries are $-H'_{ij} \geq 0$ and its diagonal entries are $a - H'_{ii} \geq 0$. Products of entrywise nonnegative matrices remain entrywise nonnegative. Because aI commutes with B,

$$e^{-\beta H'} = e^{-\beta a} e^{\beta B} = e^{-\beta a} \sum_{n=0}^{\infty} \frac{\beta^n B^n}{n!}.$$

For $\beta \geq 0$, all coefficients and the scalar factor are nonnegative. In any submultiplicative matrix norm the terms are bounded by $(\beta\|B\|)^n/n!$, whose sum converges. Thus the entrywise limit is nonnegative. At $\beta = 0$ it is the identity, so the statement is nonnegativity, not strict positivity.

An efficiency claim needs a specified sampling procedure and its cost, including relevant mixing or convergence behavior. Observable estimates also require their

preserved numerators and variance or error analysis. Matrix-entry positivity alone supplies none of these guarantees.

Credit: correct shift and matrix powers 3; series identity and coefficients 3; convergence and beta-zero case 2; computational boundary 2.

Q8. Two checks, then one (8 points)

Call	Recorded candidate checks	Status	Witness	Remaining interval
Initial budget 2	`0:000 false; 1:001 false`	`UNKNOWN`	None	$[2, 8)$
Resume budget 1	`2:010 true`	`GAUGE_VERIFIED`	Gauge 010	$[3, 8)$

Candidate 000 fails the first edge. Candidate 001 also fails it; it additionally disagrees with the closing parity. Candidate 010 satisfies every edge equation. The resumed call starts at two and ends at cursor three.

Verification of the second receipt requires the first UNKNOWN receipt, bound to the same graph and key. Its earlier checks are replayed, and the chain must begin at zero and reach the second start index without gaps. The second segment is then checked, including its witness and digest. A correct digest alone does not establish prior coverage.

Credit: initial trace 3; resumed trace 3; full prior-chain obligation 2.

Q9. Continuation attacks (6 points)

Starting at two without prior receipts is refused: the verified prefix cursor is zero. A caller-supplied count or frontier cannot supply coverage. Flipping a verdict and rehashing is rejected when replay disagrees with that verdict, even though the new digest matches the forged record.

Changing the basis changes the full presentation digest, so the old receipt cannot certify continuation under the new key. Identical Hamiltonian bytes are insufficient. The implementation also binds graph bytes and checks checker version, transformation grammar, segment order, candidate bits, verdicts, and frontier bounds.

A valid contiguous UNKNOWN chain beginning at zero supplies the prior coverage. Retrieval compares the complete declared key but returns a proposal; recheck its bits against the current graph. Neither successful retrieval nor gauge verification proves that the declared physical adapter preserves the Hamiltonian, measure, or observables.

Credit: missing prefix 2; rehashed false verdict 2; changed context and retrieval/adapter boundary 2.

Q10. Full batch and zero budget (6 points)

From zero with budget eight, the implementation completes all eight checks in its first batch. The valid candidates are indices two (010) and five (101); it selects the first in canonical order, 010. The status is `GAUGE_VERIFIED`, cursor eight, and suffix $[8, 8)$, an empty interval. Finding a witness at index two does not make the cursor three for this call.

Workers one and four produce the same ordered checks, chosen witness, and receipt. Scheduling does not enter the receipt's mathematical content. Q8's smaller budgets limit its first batch to two checks and its next to one, so its successful cursor is three.

An initial zero-budget call on this nonempty candidate space records no checks, has no witness, returns UNKNOWN, and leaves $[0, 8)$. It establishes no negative result and supplies no advancing prefix for continuation.

Credit: witness, checks, and cursor 3; worker independence and Q8 contrast 2; zero-budget limit 1.

Q11. Pairing (4 points)

On a paired subset P, the map must take values in P and satisfy $I(I(x)) = x$, $I(x) \neq x$, opposite signs, and equal magnitudes. These give disjoint canceling two-element orbits; any unpaired survivors must be assessed separately.

For $+2, -1, -1$, the total is zero but each negative magnitude is one and the only positive magnitude is two. No required partner exists. Regrouping both negative terms into a single contribution changes the carrier and permitted operation.

For real finite beta and Hermitian H on a nonzero finite-dimensional space, $\operatorname{Tr} e^{-\beta H} = \sum_j e^{-\beta \lambda_j} > 0$. Pairing every term of an absolutely convergent expansion would instead make its sum zero. Absolute convergence is needed to regroup infinitely many terms this way.

Credit: map obligations 1; counterexample and regrouping boundary 2; positive trace with convergence condition 1.

Q12. Target path (8 points)

The modes are $(\uparrow 0, \uparrow 1, \uparrow 2, \downarrow 0, \downarrow 1, \downarrow 2)$. Using zero-based mode indices:

Hop	Changed modes	Occupied modes strictly between	Sign
`110;111 -> 101;111`	1, 2	0	+1

Hop	Changed modes	Occupied modes strictly between	Sign
`101;111 -> 011;111`	0, 1	0	$+1$
`011;111 -> 110;111`	0, 2	1	-1

Each state has two doubly occupied sites, hence energy $2 \cdot 2 = 4$. Including the final root again, the dwell times sum to β; the integrand is the constant $e^{-4\beta} = e^{-1}$. The ordered three-jump simplex has volume $(1/4)^3/3! = 1/384$. The sign product is negative, so the weight is $-e^{-1}/384$.

There are 64 occupation states, but arbitrarily long closed hopping sequences are possible. Repeating a nonzero two-hop out-and-back loop already supplies unbounded lengths. Integrating times does not identify these different state sequences.

Credit: hopping signs 3; energies and dwell-time integrand 2; simplex and signed weight 2; finite states versus unbounded paths 1.

Q13. All lengths (8 points)

For an n-hop sequence, A is its positive time integral; each nonzero hopping magnitude is one.

Length	Bound or exclusion	Reason
0	$A \geq e^{-3/2}$	$E \leq 6$, beta $1/4$
1	No closed sequence	No diagonal hopping
2	$A \geq e^{-3/2}/32$	Global energy bound and $\beta^2/2! = 1/32$
3, positive	$A \geq e^{-1/2}/384$	Enumerated positive paths have $E \leq 2$
$n \geq 4$	$A \leq \beta^n/n! \leq 1/6144$	$E \geq 0$; ratio $\beta/(n+1) < 1$

The target therefore satisfies

$$\frac{1}{6144} < \frac{e^{-1}}{384} < \frac{e^{-1/2}}{384}, \qquad \frac{e^{-1}}{384} < \frac{e^{-3/2}}{32} < e^{-3/2}.$$

The first lower comparison is $e^{-1} > 1/16$; the short-path comparisons follow from the stated strict rational enclosures, rather than rounded decimal evaluations.

For $x \in [0, 1]$, successive magnitudes in $\sum_{k \geq 0} (-x)^k/k!$ decrease and tend to zero. Odd truncations bound e^{-x} below; even truncations bound it above. The checker encloses e^{-1}, $e^{-1/2}$, and $e^{-3/4}$. Its positive lower bound for the last can

be squared to bound $e^{-3/2}$. Sufficiently tight rational bounds establish the strict comparisons by fraction arithmetic.

This covers every length and excludes an individual positive path of equal magnitude. It combines finite enumeration and analytic bounds. The bound 1/6144 concerns one sequence; multiplying by a sequence count or proving convergence is required for a total-tail estimate. Time-labelled maps, collective regroupings, other bases, and other estimators require different arguments.

Credit: all five length classes and reasons 4; rational strict comparisons and decreasing long bound 2; individual-path and representation boundaries 2.

Q14. Audit (6 points)

One acceptable exact claim is: at the specified mode order and parameters, the designated negative integrated rooted path has no positive individual path of equal magnitude. Evidence is the Section 6 bounds plus `code/exact_fock_gap.py` and `code/tests/test_exact_fock_gap.py::test_exact_gap_covers_every_len`. The execution receipt records what actually ran; neither test passage nor the path obstruction classifies every physical representation.

Proposed conclusion	Correct assessment
Two failures prove no gauge	UNKNOWN with the untested suffix; complete coverage or a directly verified odd walk is missing
The gap excludes every sign-free representation	It excludes the stated individual integrated-path pairing, under the fixed parameters
The manifest verifies mathematics	It checks hashes and file census; proofs and checkers address mathematical claims
Fallback verifies the native executable	It checks the Python route; native discovery may skip until a binary is built and exercised

Credit: exact claim, domain, and named evidence 2; one point per corrected conclusion 4. Retain the scoped result rather than demanding evidence for a larger, unclaimed physical classification.

Landau theoretical minimum: PFP II

AuthorityEffect: None. The Apartment 50 preface is fictional examination staging. Use the study guide to prepare and keep the separate instructor key closed during an attempt. The manuscript defines the claims.

This is a written and oral problem set, not a historical Landau examination. Show equations, witnesses, and the scope of each conclusion. Use exact integers and fractions; numerical plots or a solver's verdict alone do not replace the requested argument.

Common instance and conventions

For Q1-Q4 and Q8-Q10, the ordered vertices are `(a,b,c)` and the undirected edges are indexed as follows:

Edge ID	Endpoints	Parity bit
0	a, b	1
1	b, c	1
2	c, a	0

The gauge equation is (_u _v=c_e). A walk step is `(edge_id, departure, arrival)`. Multiple edges and self-loops are allowed. An arrow in a walk records traversal; it does not make the edge directed or attach a face.

The search uses all (2^3=8) assignments in zero-based lexicographic order: `000,001,010,011,100,101,110,111`. Intervals are half-open: ([s,t)) includes (s) and excludes (t). Root-fixing in a proof does not change this implemented search space.

Level 0: Declare the question

Q1. Carrier and move grammar (6 points). Specify the carrier, coefficient system, allowed transformation, and acceptance equation for the common instance. What do bits zero and one require? Explain why a vertex name, a bracket-wire encoding, or an unfilled graph drawing supplies no extra mathematical structure.

Level 1: Produce and attack a witness

Q2. Gauge and its freedom (6 points). Find a gauge with (_a=0) and check all three equations. Find every gauge without that root condition. Explain what changes if a second connected component is added.

Q3. Three minimal failures (6 points). Change edge 2 to parity one. Give an indexed odd closed walk and prove that a gauge is impossible. Then give

an obstruction on one vertex and on two vertices joined by two parallel edges. State the parity bits in each case. Explain why forbidding loops would change the declared input grammar.

Level 2: Prove the criterion and respect coefficients

Q4. Same displayed numbers, different coefficients (8 points). Instead interpret the reports ((1,1,0)) as rational numbers on the directed cycle (a→b→c→a), with an exact potential requiring (r_{uv}=p_v-p_u). Compute the rational cycle sum and the binary parity. Does a rational potential exist? Find the minimum uniform (^1) change to the three reports needed to make them exact, with both a lower bound and an attaining repair. Do not import a filled triangle.

Q5. Both directions of balance (10 points). Prove that a finite graph admits a gauge exactly when every closed walk has parity zero. Handle disconnected components, loops, and parallel edges. Explain how a spanning forest produces either witness in (O(|V|+|E|)), and why that algorithm is the appropriate baseline for this grammar.

Level 3: Transfer the witness to an exact matrix

Q6. Work the adapter (8 points). For

$$H = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}, \qquad D = \operatorname{diag}(1, -1, 1),$$

derive the three graph bits and compute (DHD). Explain how zero off-diagonal entries are treated. Give the input-contract response to a float entry, a nonsquare matrix, and a nonsymmetric matrix. Does the all-positive triangle obstruct an unrestricted global eigenbasis?

Q7. Positivity without an efficiency claim (10 points). Let (H') be a finite real symmetric stoquastic matrix. For (0), prove that (e^{- H'}) is entrywise nonnegative by choosing (B=aI-H'). State why the series converges, what happens at (=0), and what additional evidence is needed before claiming efficient sampling or low-variance estimates for observables.

Level 4: Account for every tested candidate

Q8. Exact frontier trace (8 points). The common instance receives budget two from cursor zero. List the tested assignments and their verdicts, the receipt status, the witness field, and the remaining interval. Resume with budget one and the first receipt. Give the new checks, status, witness, and cursor. What does checking that second receipt require besides its own digest?

Q9. Adversarial continuation (6 points). Explain the result of each attempt: starting at index two with no prior receipts; flipping a recorded verdict

and recomputing its digest; reusing an old receipt after changing the declared basis while retaining the Hamiltonian bytes. Name the evidence that makes a valid continuation possible. Explain why an exact retrieval key still does not certify a retrieved gauge or a physical adapter.

Q10. Batches are part of the trace (6 points). Run the common instance from zero with budget eight. The implementation checks fixed batches of at most eight and finishes a batch before stopping. Which witness is selected, how many checks are recorded, and what is the cursor? Would workers one and four change these answers? Contrast this with Q8 and state what a zero-budget initial call establishes.

Level 5: Check cancellation, not just the total

Q11. Pairing obligations (4 points). State the four requirements for a sign-reversing equal-magnitude pairing on its paired subset. Explain why the carrier with contributions (+2,-1,-1) has nonnegative total but no positive equal-magnitude partner for either negative term. Does allowing a collective regrouping ask the same question? Why can an absolutely convergent thermal expansion of a nonzero finite-dimensional Hermitian system not have every term canceled in such pairs?

Level 6: Close the unbounded length argument

Q12. Derive the target weight (8 points). Use the six-mode order, parameters, and hopping-sign rule of manuscript Section 6 for

$$110;111 \longrightarrow 101;111 \longrightarrow 011;111 \longrightarrow 110;111.$$

Compute the three hopping signs, each visited energy, the simplex volume, and the signed integrated weight. Explain why 64 occupation states do not make the carrier of all rooted closed hopping sequences finite.

Q13. Exclude a partner at every length (8 points). Use the exact finite facts: 192 rooted three-hop sequences, 96 positive and 96 negative; positive three-hop paths visit energies at most two; all energies lie in ([0,6]); hops have magnitude one; and no one-hop closure exists. Derive the bounds for lengths (0,1,2,3, 4) and the strict gap around (W_ =e^{-1}/384). Explain how alternating rational partial sums certify the comparisons, why the long-path bound decreases, and why it bounds one path rather than the total tail.

Level 7: Defend the publication claim

Q14. Evidence audit (6 points). Write a short audit with one exact claim, its declared domain, a relevant package test or checker, and the limit of that evidence. Address these proposed conclusions:

- "Two failures prove there is no gauge."

- "The path gap proves that this Hamiltonian has no sign-free representation."
- "The manifest verifies the mathematics."
- "A Python fallback test verifies the native executable."

State the narrower conclusion, missing evidence, or refusal in each case. Do not require a general physical adapter as a prerequisite for the already scoped mathematical result.

Scoring and oral defense

The points sum to 100. The key allocates partial credit to calculations and arguments; a bare correct label receives no calculation or proof credit. A score of at least 80 with no unresolved scope error is the instructional competence target, not a validated learner metric.

After marking, defend three answers orally: Q5's sufficiency proof, Q8's continuation, and Q13's all-length gap. Correct any promotion of UNKNOWN to an obstruction, of a digest to truth, or of a restricted obstruction to a general physical classification before recording mastery.